Filter-matched rate calibration for variable-rate PING training

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Abstract

Poisson-encoded images are transformed by synaptic filtering, membrane integration, and finite observation windows before reaching a neural decoder. We characterized this transformation by passing MNIST pixel spike trains through the target AMPA and non-spiking conductance-based membrane. Sparse inputs produced zero-dominated, strongly skewed shot-noise features; with increasing spike count, overlapping responses formed smoother, approximately Gaussian distributions. A linear operating-point model reproduced the ordering of empirical means but overpredicted both feature mean and variability, showing that stationary linearization did not replace direct finite-window simulation. We therefore trained matched nonlinear and linear decoders on freshly simulated features across 0.01–25 Hz. On the official MNIST test set, nonlinear accuracy was reliably above chance from 0.01 Hz and reliably exceeded 50% from 0.5 Hz. The 0.5 Hz practical floor was unchanged across 0.6, 1.2, and 2.4 µS probes, supporting 0.5–25 Hz for later variable-rate PING training.

Purpose and scope

The question is:

> At a given encoding rate, does the temporally filtered image still contain enough information for an ANN to recognize the digit?

The target PING architecture has 784 pixel channels feeding conductance-based excitatory cells. Its fixed synaptic and membrane equations define the features, without trained input, recurrent, or output weights. Matched linear and nonlinear decoders therefore measure decoder-relative accessibility, not an absolute information boundary or PING accuracy.

Methods

1. **Prepared the dataset partitions.** MNIST provides an official 60,000-image training set and a separate 10,000-image test set. We split the official training set into a 55,000-image training subset, used to fit decoder parameters, and a 5,000-image validation subset, evaluated after every epoch to select the best checkpoint. Validation also selected among the linear decoder's three L2 weight-decay values. Only after these choices and the evaluation protocol were frozen did we load the official test set for the psychometric curves and exploratory rate-range decision.

2. **Generated and validated filter-matched pixel features.**

   Figure 1 illustrates the single-pixel feature dynamics.

Every image used a 200 ms presentation and 0.1 ms timestep. Decoder training and evaluation treated 0.01, 0.05, 0.1, 0.25, 0.5, 0.75, 1, 1.5, 2, 2.5, 3, 4, 5, 10, 25 Hz as a uniform categorical grid. For separate characterization, we constructed an **empirical response table**: a multidimensional lookup array of simulated single-pixel features z, indexed by random seed, probe conductance, encoding rate, uint8 pixel intensity, and stochastic draw. The table was used only to estimate and plot the response mean and variance at each condition, compare table sampling with fresh direct simulation, and assess the analytical approximation. It never generated decoder inputs. It covered the 0.25, 0.5, 0.75, 1, 1.5, 2, 2.5, 3, 4, 5, 10, 25 Hz characterization grid, while decoder training and evaluation used fresh direct simulations at all 15 rates. Seeds 42, 43, 44 defined independent feature and table-generation runs.

Each pixel drove an independent AMPA synapse and uncoupled non-spiking excitatory membrane. This captured synaptic decay, conductance-dependent integration, and within-window timing, but included no threshold, reset, recurrence, inhibition, shared noise, or trained synaptic weights.

The complete feature path is

$$S_i(t) \to g_i^{\text{pix}}(t) \to v_i(t) \to z_i \to \text{ANN}. \quad (1)$$

Here S_i, $g^{\text{pix}}{}_i$, v_i, and z_i are pixel i's spike train, AMPA conductance, subthreshold voltage, and time-averaged feature; t is time and ANN is the classifier.

All dynamical equations used milliseconds. We approximated independent Poisson input in discrete time. At each timestep, pixel i could emit at most one event, drawn according to

$$S_i(t) \sim \text{Bernoulli}\left(rx_i\Delta\frac{t}{1000}\right). \quad (2)$$

Here r is the reported maximum-pixel encoding rate in spikes/s, x_i is normalized pixel intensity, division by 1000 reconciles seconds with the millisecond timestep Δt, and Bernoulli is an independent binary draw. Pixel i therefore had reported rate rx_i Hz.

We updated the AMPA conductance using

$$g_i^{\text{pix}}(t) = \beta_{\text{AMPA}} g_i^{\text{pix}}(t-1) + w_{\text{probe}} S_i(t). \quad (3)$$

$$\beta_{\text{AMPA}} = \exp\left(-\frac{\Delta t}{\tau_{\text{AMPA}}}\right). \quad (4)$$

Here β_{AMPA} is one-timestep conductance retention, $\tau_{\text{AMPA}} = 2$ ms is the AMPA decay time, w_{probe} is the AMPA conductance increment produced by one pixel spike. Other symbols follow Equation 2. The primary 1.2 µS probe is the target PING network architecture's nominal mean initial pixel-to-excitatory conductance; 0.6 and 2.4 µS test half and double scale for comparison.

We integrated the membrane without thresholding or resetting:

$$C_E\frac{dv_i}{dt} = g_{\text{L,E}}(E_L - v_i) + g_i^{\text{pix}}(t)(E_e - v_i). \quad (5)$$

$$\tau_{\mathrm{eff}}, i(t) = \frac{C_E}{g_{\mathrm{L,E}} + g_i^{\mathrm{pix}}(t)}. \quad (6)$$

Here C_E is excitatory capacitance; $g_{L,E}$ is leak conductance; E_L and E_e are leak and AMPA reversal potentials; and $\tau_{eff,i}$ is the instantaneous conductance-dependent time constant. Other symbols follow Equation 1.

Voltage began at E_L and conductance at zero. After one presentation,

$$z_i = \frac{1}{T} \int_0^T \left(v_i(t) - E_L\right) \mathrm{d}t. \quad (7)$$

where T is presentation duration and z_i is mean baseline-subtracted voltage over the complete window, not terminal voltage. Threshold and reset were disabled, so "subthreshold" refers to the non-spiking membrane equations rather than a constraint on attained voltage. Neither the feature nor the decoder input included rate, so sparse inputs retained their weaker, noisier signal.

Figure 1 shows the spike train, conductance response, membrane response, and persistence of timing information after window averaging.

3. **Characterized the empirical response moments.**

Figures 2 and 3 show the empirical response moments and how the full response distribution changes with input rate.

The stored table contained three seeds, three conductances, the twelve 0.25, 0.5, 0.75, 1, 1.5, 2, 2.5, 3, 4, 5, 10, 25 Hz rates, all 256 uint8 intensities, and 2048 simulations per condition and seed. Figure 2 used the first 512 draws from each seed as a practical plotting subset. This subset size was neither a convergence gate nor an ANN-training requirement. It estimated the mean response

$$\hat{\mu}_z\left(x, r, w_{\mathrm{probe}}\right) = \frac{1}{K} \sum_{k=1}^{K} z^k. \quad (8)$$

and response variance

$$\hat{\sigma}_z^2\left(x, r, w_{\mathrm{probe}}\right) = \frac{1}{K-1} \sum_{k=1}^{K} \left(z^k - \hat{\mu}_z\right)^2. \quad (9)$$

Here x, r, and w_{probe} are pixel-intensity, rate, and probe conductance; $z^{(k)}$ is draw k; K is the draw count.

Figure 2 plots these estimated response means and SDs across the calibrated rate–intensity conditions.

Figure 3 plots all 2048 simulations from each of the three seeds at maximum pixel intensity, the nominal 1.2 µS probe, and representative low, intermediate, and high rates. A Gaussian with the same empirical mean and SD was overlaid at each rate to assess distribution shape.

In contrast to this diagnostic lookup array, ANN decoder presentation n used a newly generated spike train and direct synapse–membrane simulation:

$$S_i^n(t) \to g_i^{\text{pix,n}(t)} \to v_i^n(t) \to z_i^n \to \text{ANN}. \quad (10)$$

Repetition sampled the full response distribution at each condition without an intermediate noise model.

4. **Derived the analytical linear-filter response.**

Figures 4 and 5 show the analytical transfer functions and the analytical moments over input drive, respectively.

We calculated a local linear approximation to the synapse, membrane, and finite averaging window. This diagnostic was not used to select a training range. Appendix A derives the equations in this section.

For reported encoding rate r and normalized intensity x, the analytical rate was $\lambda = rx$ spikes/s. Its stationary mean conductance is

$$\overline{g}_\lambda = \lambda w_{\text{probe}} \frac{\tau_{\text{AMPA}}}{1000}. \quad (11)$$

and the deterministic operating-point voltage is

$$\overline{v}_\lambda = \frac{g_{\text{L,E}} E_L + \overline{g}_\lambda E_e}{g_{\text{L,E}} + \overline{g}_\lambda}. \quad (12)$$

At this operating point, voltage is constant across the presentation, so applying the baseline subtraction and time average in Equation 7 gives the analytical mean feature

$$\mu_{\text{linear}(z)} = \overline{v}_\lambda - E_L. \quad (12a)$$

Here $\overline{g}_\lambda$ is stationary mean conductance, $\overline{v}_\lambda$ is the voltage obtained by replacing the fluctuating conductance with $\overline{g}_\lambda$, and μ_{linear} (z) is the predicted mean of the final feature z. Other symbols follow Equations 3–7.

Linearizing the synapse and membrane around that operating point gives

$$G_{\lambda(\omega)} = \frac{w_{\text{probe}}}{i\omega + \frac{1}{\tau_{\text{AMPA}}}} \cdot \frac{E_e - \overline{v}_\lambda}{i\omega C_E + g_{\text{L,E}} + \overline{g}_\lambda}. \quad (13)$$

Here G_λ (ω) is the local synapse-plus-membrane transfer function; ω is angular frequency in rad/ms.

The finite averaging window contributes

$$A_{T(\omega)} = \frac{1 - \exp(-i\omega T)}{i\omega T}. \quad (14)$$

$$H_{\lambda(\omega)} = A_{T(\omega)} G_{\lambda(\omega)}. \quad (15)$$

Here A_T (ω) is the duration-T averaging response; H_λ (ω) is the complete response from input fluctuation to averaged feature and T is presentation duration in ms.

Figure 4, Panel A plots G_λ from Equation 13, and Panel B plots the complete window-averaged H_λ from Equation 15.

Appendix A.4 derives the input spectrum from the discrete Bernoulli encoder, its continuous-time Poisson limit, and the resulting autocovariance. The centred ideal Poisson input has the constant, or white, two-sided spectrum

$$S_{\mathrm{in}(\omega)} = \frac{\lambda}{1000}. \quad (16)$$

$$S_{z(\omega)} = \left|H_{\lambda(\omega)}\right|^2 S_{\mathrm{in}(\omega)}. \quad (17)$$

$$\mathrm{Var}_{\mathrm{linear}(z)} = \frac{1}{2\pi}\int_{-\infty}^{\infty} \left|H_{\lambda(\omega)}\right|^2 S_{\mathrm{in}(\omega)}\,\mathrm{d}\omega. \quad (18)$$

Here λ is in spikes/s and division by 1000 accounts for the millisecond time base; S_{in} (ω) and S_z (ω) are input and predicted output power spectral densities; $|H_\lambda (\omega)|^2$ is transmitted noise power, and Var_{linear} (z) is predicted feature variance.

At every rate, intensity, and conductance, we set $\lambda = rx$, selected the corresponding operating-point model, and compared its predictions with the empirical response table without new simulation. Analytical mean was Equation 12a ($\mu_{\mathrm{linear}(z)}$); analytical SD was the square root of Equation 18 ($\mathrm{Var}_{\mathrm{linear}(z)}$). Figure 5 plots the analytical mean and SD over the same expected-input-spike axis as Figure 2, with the empirical values overlaid.

5. **Constructed and compared complete MNIST feature images.**

   Figure 6 compares the resulting complete feature images.

   For each uint8 pixel, the diagnostic sampler used its exact 0–255 intensity index and an empirical draw with independent deterministic pixel and image streams. A fresh direct simulation provided the comparison. This used training images 0–15, never validation or test images, and covered all 28 × 28 pixels, eight replicates, three conductances, and representative sparse, intermediate, and upper-grid conditions at 0.25, 3, and 25 Hz.

   Figure 6 places the original MNIST image, an empirical-table sample, and a fresh direct simulation side by side.

6. **Trained mixed-rate decoders.**

   Figure 7 shows the validation histories used for epoch selection.

   Each simulated image produced a feature vector $\boldsymbol{z} \in \mathbb{R}^{784}$, containing one time-averaged membrane-voltage displacement per pixel. The primary ANN transformed this vector through a fully connected 1,024-unit hidden layer,

$$\boldsymbol{h} = \mathrm{ReLU}(W_h\boldsymbol{z} + \boldsymbol{b}_h), \quad \boldsymbol{s} = W_o\boldsymbol{h} + \boldsymbol{b}_o,$$

   where $\boldsymbol{h}$ is hidden activation, $\boldsymbol{s} \in \mathbb{R}^{10}$ contains one logit per digit, and elementwise $\mathrm{ReLU}(x) = \max(0, x)$ retains positive inputs and zeros negative ones. Both weight matrices and bias vectors were trainable; prediction used $\arg\max_c s_c$. The 784–1,024–10 network had no convolution, recurrence, normalization, dropout, or pretrained weights.

   The matched linear decoder omitted the hidden layer and computed

$$\boldsymbol{s} = W_{\mathrm{lin}}\boldsymbol{z} + \boldsymbol{b}_{\mathrm{lin}}.$$

Its 784 × 10 weights and ten biases were trainable. “Linear” described the feature-to-logit map, not fitting: both models used backpropagation. Linear performance measured directly accessible digit evidence; the ANN advantage measured the benefit of its nonlinear representation. Neither received rate.

Both models were fitted with ten-class cross-entropy. For logits $\boldsymbol{s}$ and true digit label y, the loss for one image was

$$\mathcal{L}_{\mathrm{CE}(\boldsymbol{s},y)} = -\log\left(\frac{\exp(s_y)}{\sum_{c=0}^{9}}\exp(s_c)\right).$$

Backpropagation minimized this penalty for low correct-class probability.

Each image presentation independently sampled one of the 15 registered rates with probability 1/15. That maximum rate applied to the complete image, while each pixel was scaled by its intensity. Each of the 55,000 training images appeared once per epoch for 15 epochs, with a newly sampled rate and fresh spike trains on every presentation. The same 5,000 validation images were evaluated after every epoch using new reproducibly seeded rate assignments and spike trains.

Within a training run, the ANN and linear models received the same directly simulated feature batches. Adam used learning rate 0.001 and batch size 256. The ANN used no weight decay, and validation accuracy selected its best epoch. We also trained three otherwise identical linear decoders with L2 weight decays 10^{-5}, 10^{-4}, and 10^{-3}. The candidate with the best validation accuracy was selected. The nonlinear architecture and absence of weight decay were fixed rather than tuned. Test accuracy played no role in these choices.

Seeds 42, 43, 44 controlled parameter initialization and stochastic training and validation features, so they represented complete training replicates. Every conductance and seed had a separate decoder; runs were not pooled during training.

Figure 7 plots the validation histories from which the best epochs were selected.

7. **Evaluated frozen official-test psychometric curves and selected the rate range.**

   Figure 8 shows the official-test psychometric curves and derived thresholds.

   After validation had fixed the best epochs and linear weight decay, we loaded the same official 10,000 MNIST test images for every condition. At each rate and conductance, we encoded every image three times with newly simulated spike trains. Within one rate, conductance, and encoding draw, all three trained decoder seeds received exactly the same feature vectors. At the nominal 1.2 µS conductance, the selected ANN and linear decoder also shared those features. Different conductances used different spike-train realizations, and discarded linear weight-decay candidates were not tested. Accuracy at each rate was the mean across test images, the three encodings, and the three trained decoders. The primary curve was

   $$A_r(r) = P(\text{correct} \mid r, \text{mixed-rate nonlinear decoder}). \quad (19)$$

   Here A_r is official-test nonlinear-ANN accuracy, P is correct-classification probability, and r is rate. The linear decoder was diagnostic.

We estimated uncertainty separately at every tested rate and conductance; rates were never mixed in this calculation. For each condition, we created 2,000 alternative versions of the recorded evaluation. One version was formed by selecting 10,000 entries from the list of test images, three from the list of encoding draws, and three from the list of trained decoders. After each selection the chosen item remained available, so an image, draw, or decoder could appear several times while another could be absent. This is sampling "with replacement." We calculated accuracy for each alternative version. L_r was the fifth percentile of those 2,000 accuracies: after sorting them, 95% were at least L_r. The decoder-relative edge was

$$r_{\text{decode}} = \text{lowest} \;\; r \in \mathcal{R} \;\; \text{satisfying} \;\; L_r(r) > \frac{1}{N_{\text{class}}}. \quad (20)$$

The practical training floor was

$$r_{\text{train}} = \text{lowest} \;\; r \in \mathcal{R} \;\; \text{satisfying} \;\; L_r(r) \geq a_{\text{use}}. \quad (21)$$

Here $\mathcal{R}$ is the tested grid, N_{class} the class count, r_{decode} the lowest rate reliably above chance, a_{use} the 50% useful-accuracy target, and r_{train} its first reliable crossing. The 10% chance criterion corresponds to uniform guessing across ten classes. Primary thresholds used the 1.2 µS ANN; the linear decoder and other conductances measured sensitivity. The 784-to-10 decoder without a hidden layer was evaluated on the test set only at the nominal 1.2 µS conductance, not at 0.6 or 2.4 µS. Accuracies were measured only at the 15 listed rates; no values between adjacent rates were estimated or treated as data. We used the resulting r_{train} as the lower bound of the later PING training range and the highest tested rate, 25 Hz, as its upper bound. The 50% criterion was a pragmatic screen for substantial digit information, not a theoretical information boundary.

Figure 8 plots the official-test psychometric curves and the thresholds obtained from this procedure.

Results

Filter-matched feature generation

We first asked whether the feature retained the timing imposed by the AMPA synapse and finite presentation window. A single spike was placed at different times within a 200 ms presentation, passed through the uncoupled conductance and membrane model, and reduced to the time-averaged feature z.

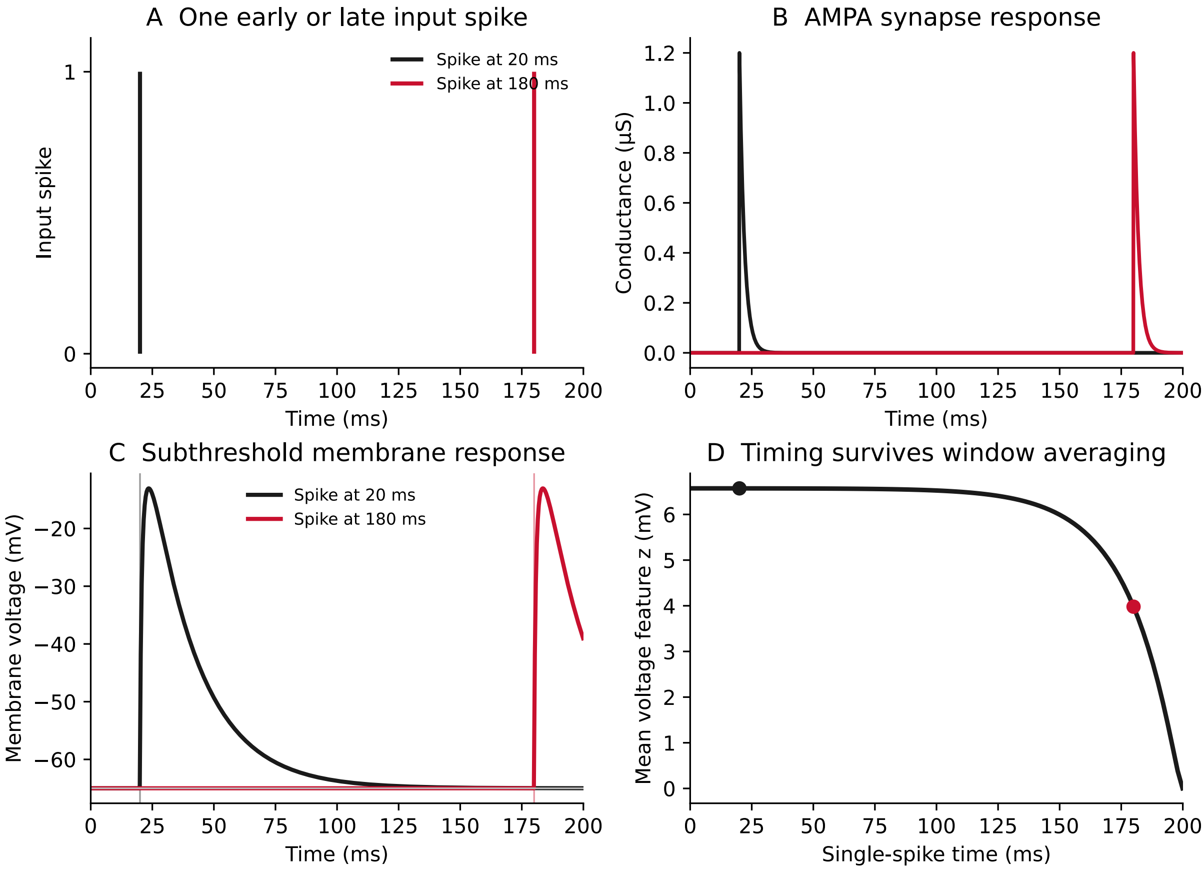

Figure 1: Finite-window timing sensitivity. Panel A places one input spike at 20 or 180 ms. Panels B and C pass those inputs through the registered AMPA synapse and subthreshold membrane. Panel D repeats the simulation across 101 single-spike times and plots the presentation-averaged feature z. Conductance and voltage relax after each spike; z falls for later spikes because the fixed 200 ms window truncates more of the response.

Empirical response moments

We next characterized the magnitude and variability of z under stochastic encoding across all 256 pixel intensities, 12 calibration rates, and three probe conductances. The stored table contained 2048 simulations for each condition and random seed. For Figure 2, we chose the first 512 simulations per seed as a smaller practical plotting subset. Combining the three independent seeds gave 1536 feature values from which we calculated one mean and sample SD for each plotted condition.

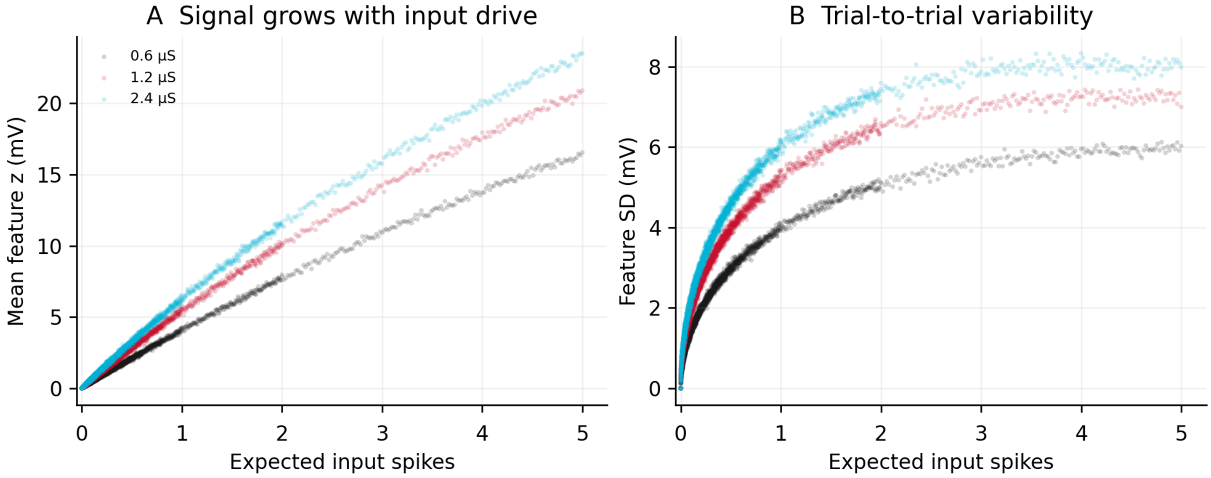

Figure 2: Mean and variability of the simulated single-pixel feature. We evaluated 9,216 conditions: 256 pixel intensities × 12 encoding rates × three probe conductances. For each condition, we selected 512 simulations from each of three independent random seeds, giving 1536 values of feature z. Each point in Panel A is the mean of those 1536 values; the corresponding point in Panel B is their standard deviation. Thus, each panel contains 9,216 points. The 512 simulations per seed were chosen as a practical descriptive subset of the available simulations, not through a statistical convergence test. The horizontal axis is the expected number of input spikes during the 200 ms presentation: encoding rate × normalized pixel intensity × 0.2 s. Colour identifies probe conductance. Mean response increases with expected spike count. Variability initially rises as spike counts and times diversify, then can plateau or fall when averaging, shunting, and voltage saturation limit additional excursions.

Distribution shape across rates

To distinguish sparse shot noise from a high-count approximation, we examined the complete recorded z distribution for a maximum-intensity pixel at the nominal 1.2 µS probe. The three rates give 0.05, 0.6, and 5 expected spikes during one 200 ms presentation, where $E[N] = \lambda\frac{T}{1000} = rx\frac{T}{1000}$.

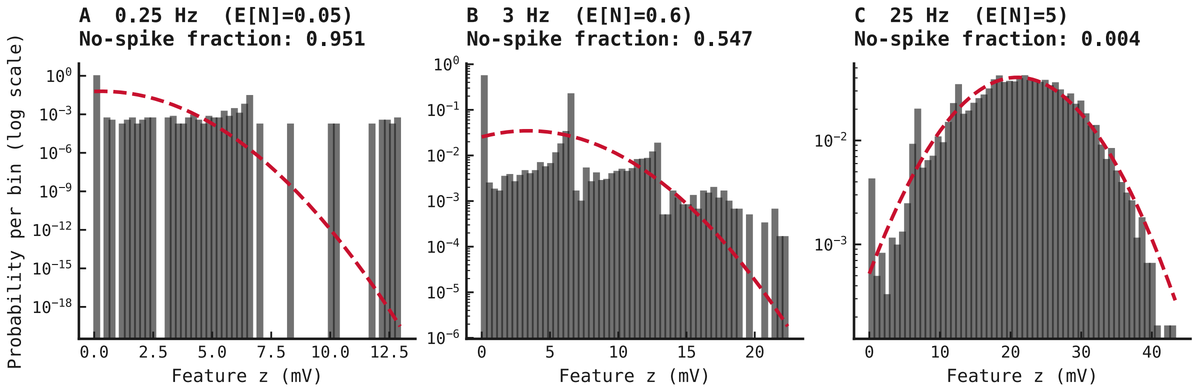

Figure 3: Transition from sparse shot noise toward a Gaussian-like feature distribution. Each panel contains 6144 recorded z values: 2048 simulations × three seeds for pixel intensity 255 and the nominal 1.2 µS probe. Bars show empirical probability per bin on a logarithmic axis; the red dashed curve is a Gaussian with the same empirical mean and SD. Panel A, at 0.25 Hz, is dominated by no-spike presentations and separated responses to rare spike counts and times. Panel B, at 3 Hz, remains strongly non-Gaussian because more than half of presentations contain no spike. Panel C, at 25 Hz, contains about five expected spikes per presentation; overlapping filtered responses form a much smoother distribution that approaches the matched Gaussian, while conductance nonlinearity and finite-window timing leave visible deviations.

Analytical linear-filter response

Analytical frequency response

We then evaluated the linearized synapse-plus-membrane response and its 200 ms window-averaged form from 0.1–200 Hz at low, intermediate, and high operating rates. These curves were calculated from Equation 13 (G_λ) and Equation 15 (H_λ), rather than from simulated traces.

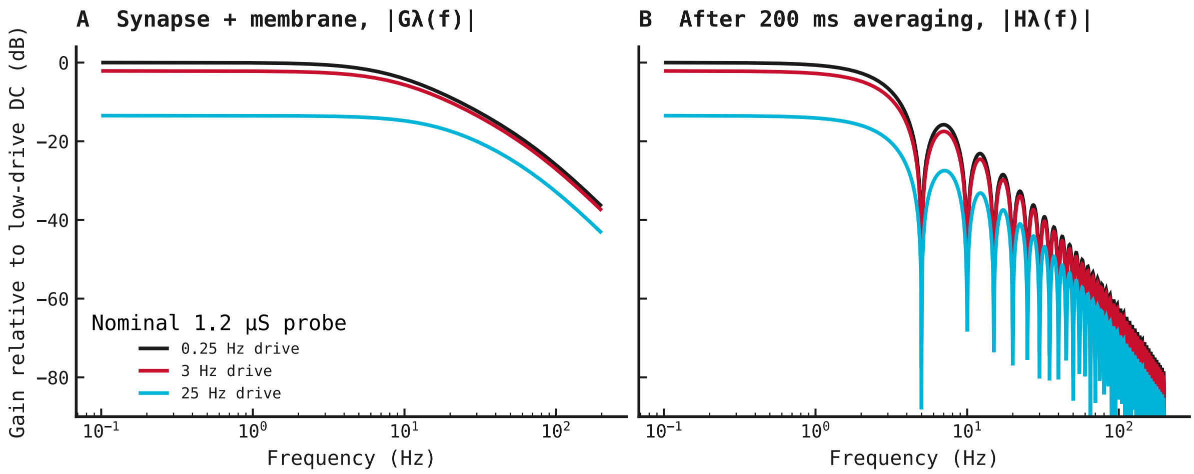

Figure 4: Analytically calculated linearized frequency response. All curves use maximum pixel intensity $x = 1$ and the nominal probe conductance $w_{\text{probe}} = 1.2$ µS. The three operating rates are therefore $\lambda = 0.25$, 3, and 25 spikes/s; each curve is the corresponding local member of the Equation 13 family. The functions contain no simulated traces. Frequency f is a small sinusoidal modulation around each operating rate. Panel A evaluates Equation 13 (G_λ) from 0.1–200 Hz, with $\omega = 2\pi \frac{f}{1000}$ rad/ms. Gain is

$$20 \log_{10} \left(\frac{\left| G_{\lambda}(2\pi \frac{f}{1000}) \right|}{\left| G_{\lambda_{\text{low}}}(0) \right|} \right),$$

relative to the 0.25 Hz DC response. Thus the black curve begins near 0 dB; higher mean conductance lowers the red and cyan DC gains. Panel B plots Equation 15 ($H_\lambda = A_T G_\lambda$) after the analytical 200 ms rectangular-window average.

Panel A combines an AMPA low-pass pole at $\frac{1}{\tau_{\text{AMPA}}}$ with a membrane pole set by effective conductance and capacitance, suppressing fast modulation. Greater drive shunts and depolarizes the membrane, reducing excitatory driving force and separating the curves.

Panel B adds the averaging magnitude. Writing the presentation duration in seconds as $T_s = \frac{200}{1000}$ s, $\left| A_{T}(2\pi \frac{f}{1000}) \right| = \left| \frac{\sin(\pi f T_s)}{\pi f T_s} \right|$. It is one at zero frequency and zero at $f = \frac{n}{T_s}$ for nonzero integer n. Integer cycles therefore cancel at 5, 10, 15 Hz, and so on. Magnitude folds alternating sinc signs into positive lobes; the two low-pass terms attenuate successive lobes into the falling envelope.

Analytical moments over input drive

We next plotted the analytical moments over the same expected-input-spike coordinate as the empirical response in Figure 2. This exposes the predicted scale and curvature before collapsing each condition into a predicted-versus- empirical point.

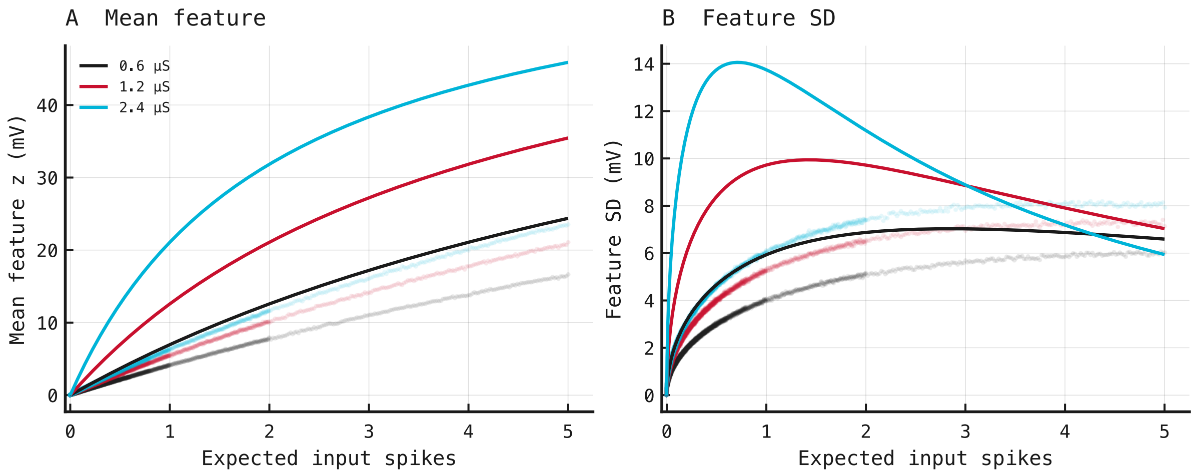

Figure 5: Analytical response moments over input drive. The horizontal axis matches Figure 2: expected input spikes are $\lambda \frac{T}{1000} = rx \frac{T}{1000}$. Panel A shows the analytical mean feature $\mu_{\text{linear}(z)}$ from Equation 12a; Panel B shows the analytical SD $\sqrt{\text{Var}_{\text{linear}(z)}}$ from Equation 18. Solid curves are analytical predictions; faint points are the corresponding empirical values from Figure 2. Black, red, and cyan denote probe conductances 0.6, 1.2, and 2.4 µS. The shared drive coordinate reveals mean overprediction directly and shows that the analytical and empirical SD maxima occur at different input drives.

The curves differ because the analytical model describes small, continuous fluctuations around a stationary mean conductance, whereas the empirical presentations contain only approximately 0–5 discrete input spikes. In this sparse regime, individual spike counts and times dominate: there is no densely sampled mean state around which the dynamics fluctuate weakly.

In Panel A, Equation 12a effectively evaluates the feature at the mean conductance, $z(E[g])$, while the empirical curve estimates the mean feature over random conductance paths, $E[z(g)]$. These differ because the conductance-to-voltage response is nonlinear and saturating, giving approximately

$$E[z(g)] < z(E[g]).$$

For example, one expected spike describes a mixture of presentations with zero, one, two, or more spikes, not one continuously distributed input. A late spike contributes little to the 200 ms feature average, whereas the stationary model spreads the mean input across the entire presentation. These effects make the analytical mean systematically too high. The discrepancy grows with probe conductance because each 2.4 µS event is a larger nonlinear perturbation, making the local approximation least credible for the cyan curve.

In Panel B, Equation 18 propagates ideal Poisson noise through the local linear filter in Equation 15. The empirical feature distribution is instead a discrete spike-count and spike-time mixture,

$$P(z) = \sum_n P(N = n) P(z \mid N = n),$$

where N is the spike count during one presentation. Within each count, different spike times produce another distribution of responses. At low drive, transitions between zero and one large spike give the analytical model a very large local response, especially at 2.4 μS, so its SD rises sharply and peaks early. The empirical membrane instead saturates during large jumps; zero-spike trials remain exactly at rest; and late spikes contribute only partially. This strongly non-Gaussian mixture cannot be represented by a single power spectrum and local gain.

With increasing drive, relative count variability decreases, timing differences average out, conductance shunts the membrane, and depolarization reduces the excitatory driving force. The analytical SD therefore falls after its early maximum. The empirical SD rises more gradually because it moves through successive spike-count mixtures rather than fluctuating around one operating point. Thus the mean mismatch primarily reflects $E[z(g)] \neq z(E[g])$ plus spike timing, while the displaced SD maximum reflects a sparse count-and-timing mixture being approximated as stationary Gaussian noise.

Complete feature images

We then tested whether independently generated single-pixel responses formed recognizable complete images. For the same MNIST training images and nominal probe, we compared the original pixels with features sampled from the empirical response table and with fresh direct simulations at representative low, intermediate, and high rates.

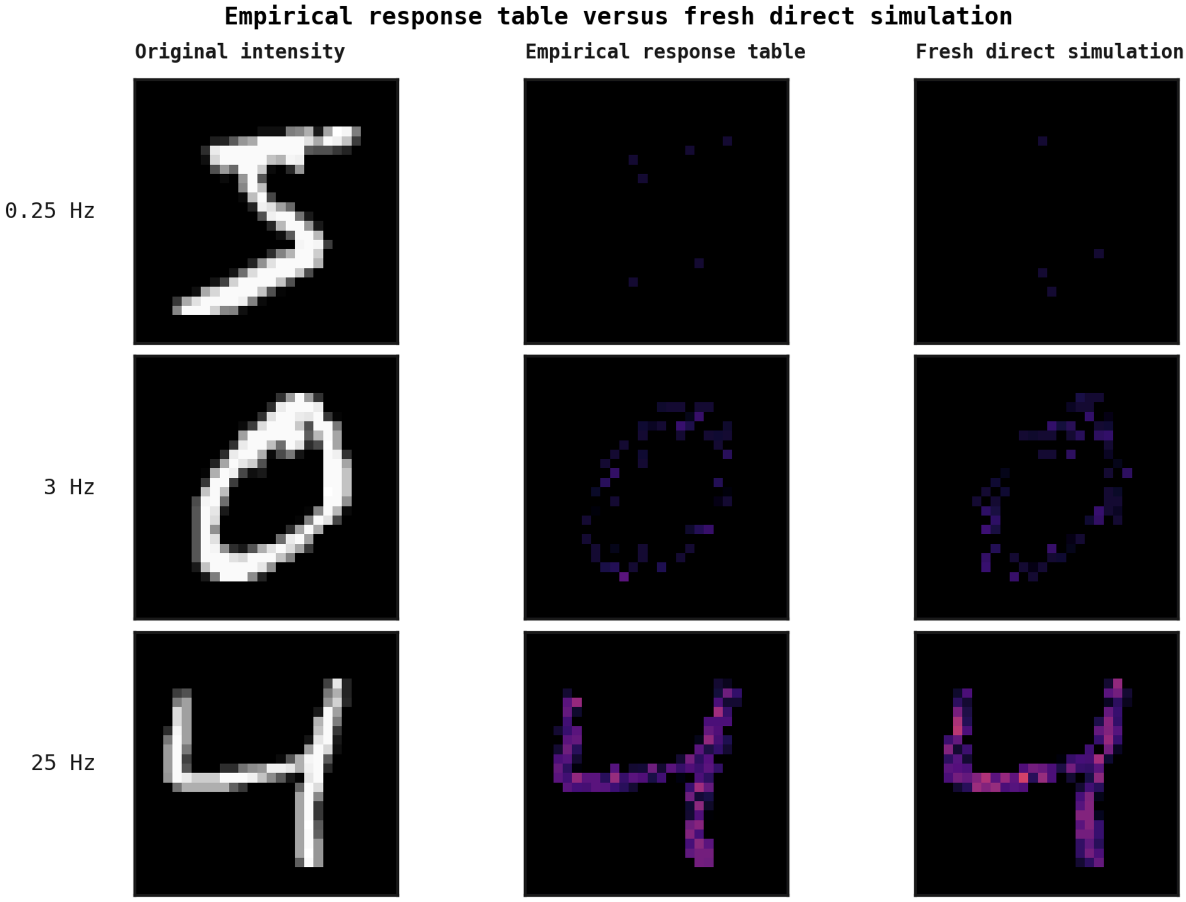

Figure 6: Training-image feature comparison. Rows show an original MNIST training image, a response-table sample, and a fresh direct simulation at 0.25, 3, and 25 Hz for the nominal 1.2 μS probe, using common 0–65 mV limits. Increasing rate reveals the intensity pattern as spatial signal grows relative to independent sampling noise.

At 0.25 Hz, median image-level mean differences between response-table and direct features were 49.68%, 47.19%, and 56.12% at 0.6, 1.2, and 2.4 µS. Median variance differences were 52.72%, 53.22%, and 56.83%, and spatial correlations were 0.236, 0.217, and 0.198.

At 3 Hz, the corresponding median mean differences were 12.83%, 11.4%, and 11.02% median variance differences were 19.29%, 19.93%, and 17.55% and spatial correlations were 0.776, 0.78, and 0.788.

At 25 Hz, median mean differences were 4.27%, 3.52%, and 4.07% median variance differences were 8.19%, 6.95%, and 7.18% and spatial correlations were 0.974, 0.977, and 0.977.

Mixed-rate decoder training

We trained the nonlinear ANN and matched linear decoder on fresh directly simulated features while sampling all 15 rates equally. After every epoch, the fixed 5,000-image validation subset was freshly encoded and evaluated; these histories selected the best epoch for each trained run.

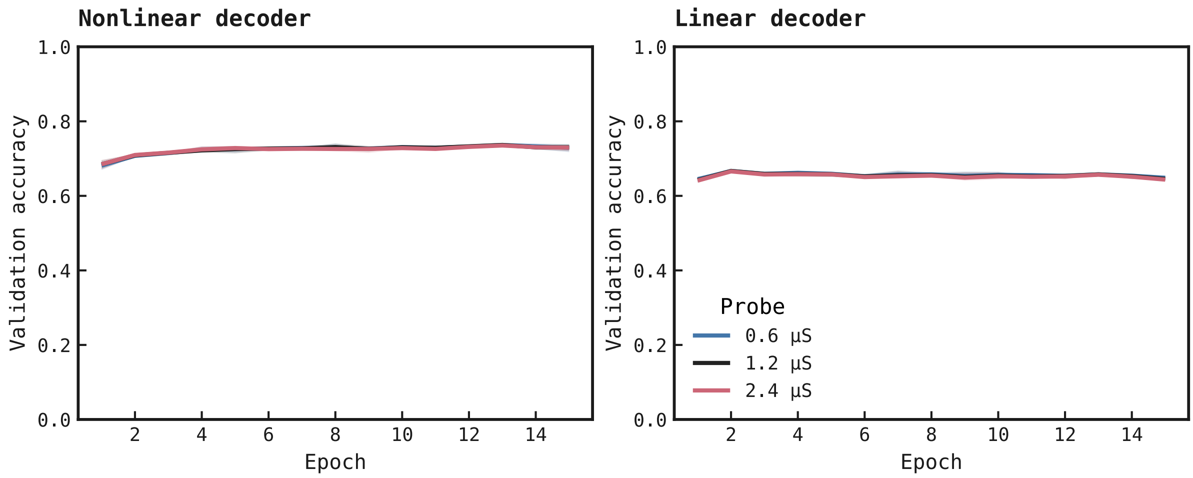

Figure 7: Mixed-rate validation histories. Each panel plots mean validation accuracy across decoder seeds, with their range shaded, under equal sampling of all 15 rates from 0.01–25 Hz and fresh direct simulations. Curves rise then flatten as class structure is acquired; conductances overlap because response scaling preserves similar spatial evidence.

At the nominal probe, seeds 42, 43, and 44 respectively selected nonlinear epochs 15, 13, and 8, with accuracies 73.9%, 73.62%, and 74.04%. The matched linear decoders reached 66.8%, 66.38%, and 66.88%.

Official-test psychometric curves and rate range

After model selection was complete, we evaluated the frozen decoders on the official 10,000-image MNIST test set at each rate. Every image was encoded three times, and accuracy was averaged across those encodings and three independently trained decoder runs before applying the above-chance and 50% practical criteria.

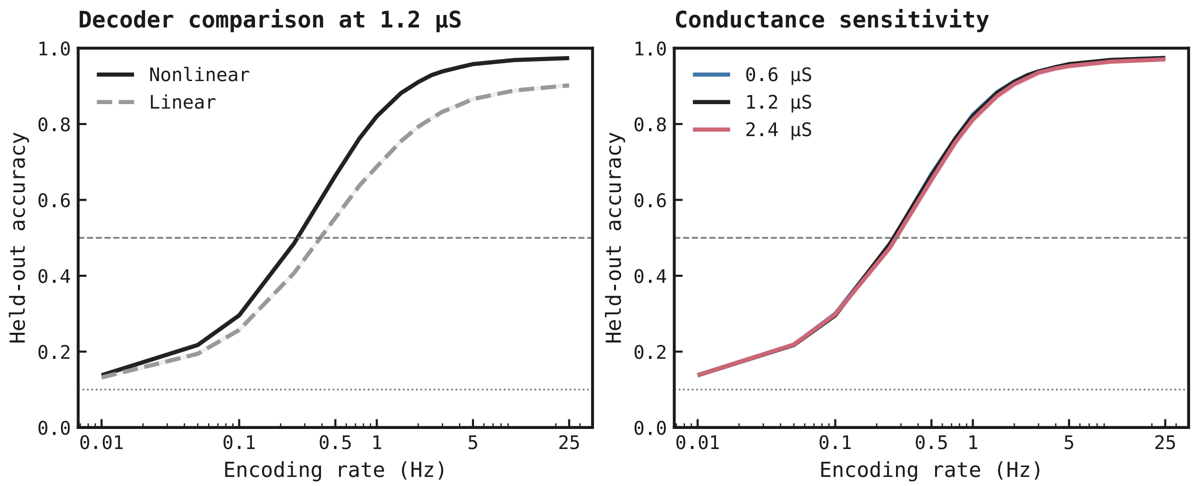

Figure 8: Official-test psychometric curves. Each point pools the official 10,000-image test partition across three fresh simulation draws and three independently trained runs. Within each rate, conductance, and draw, runs share feature vectors; different conductances use different spike draws. Accuracies, not predictions, are averaged. Bands bootstrap images, draws, and runs; horizontal rules mark 10% uniform guessing and the 50% pragmatic criterion. Accuracy rises as more pixels spike within 200 ms. Because each conductance has separately retrained decoders, overlap shows recoverability after retraining, not robustness of one fixed decoder.

At the nominal probe, accuracy was 13.77% at 0.01 Hz, 21.74% at 0.05 Hz, and 29.56% at 0.1 Hz. Their one-sided lower bounds were 13.21%, 21.14%, and 28.92%, respectively. Thus the lowest tested rate exceeded 10% chance, but the ultra-sparse range remained below the 50% practical criterion. At 0.01 Hz, a maximum-intensity pixel emitted only 0.002 expected spikes per presentation, so this result reflects rare events and decoder priors rather than a visually complete digit.

At 0.25 Hz, nominal nonlinear accuracy was 48.48%, with a 47.77% one-sided lower bound. At 0.5 Hz it was 66.39%, with a 65.64% lower bound. The nominal linear decoder also first crossed the 50% lower-bound criterion at 0.5 Hz. The reported 0.5 Hz floor is the lowest tested rate satisfying the criterion, not an estimate that the continuous crossing occurs exactly at 0.5 Hz.

Training-range decision

The nonlinear decoder was reliably above chance from 0.01 Hz, and its practical floor was 0.5 Hz. The practical floor remained 0.5 Hz at 0.6 µS, 0.5 Hz at 1.2 µS, and 0.5 Hz at 2.4 µS. We therefore recommend rates from 0.5 Hz to 25 Hz for later variable-rate PING training. This decoder-relative range is neither an absolute information limit nor a PING accuracy result. Expanded retraining and evaluation cost 5.136 USD on Modal, bringing cumulative exp080 compute to 9.43 USD.

Limitations

This measured decoder-relative accessibility, not PING performance, and each conductance used separately retrained decoders. Because the official test partition informed the exploratory rate decision, confirmation requires new held-out data. The response table omitted the three lowest rates, although direct-simulation training and test evaluation included them equally. Three training runs and three noise draws provide limited between-run and between-draw sampling.

Relation to prior work

Filtered conductance shot noise can violate fixed-time-constant models [**1**], and Gaussian approximations can miss skewed responses [**2**], supporting the empirical response table. Decoder performance is not absolute neural information [**3**], and linear and nonlinear decoders can differ [**4**], motivating a decoder-relative edge and both decoder diagnostics.

References

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Appendix A: linear-filter derivation

This appendix derives Equations 11–18 from the same AMPA synapse and subthreshold membrane used by the empirical probe. It replaces the finite Bernoulli drive with a continuous-time Poisson point process and linearizes the membrane around the mean drive at each calibration condition.

A.1 Stationary operating point

Represent pixel i's idealized spike train and synaptic conductance by

$$s_{i(t)} = \sum_k \delta(t - t_k), \quad \frac{dg_{i(t)}}{dt} = -\frac{g_{i(t)}}{\tau_{\mathrm{AMPA}}} + w_{\mathrm{probe}} s_{i(t)}. \quad \text{(A1)}$$

Here time is measured in ms; s_i (t) is a sum of Dirac impulses; t_k is spike k's time; δ is the Dirac delta; g_i (t) is AMPA conductance; $\tau_{\mathrm{AMPA}} = 2$ ms is its decay time; w_{probe} is conductance added per spike; and t is time. Equation A1 is the continuous-time counterpart of the decay-then-add update in Equation 3.

For stationary Poisson rate λ in spikes/s, the spike train contributes an expected λ/1000 events per millisecond and the mean conductance satisfies

$$0 = -\frac{\overline{g}_\lambda}{\tau_{\mathrm{AMPA}}} + w_{\mathrm{probe}} \frac{\lambda}{1000}, \quad \overline{g}_\lambda = \lambda w_{\mathrm{probe}} \frac{\tau_{\mathrm{AMPA}}}{1000}. \quad \text{(A2)}$$

Here λ is encoding rate times pixel-intensity, and $\overline{g}_\lambda$ is stationary mean conductance. The zero on the left states that the mean no longer changes with time.

To define the deterministic operating point, replace the fluctuating conductance by $\overline{g}_\lambda$ and set the membrane derivative in Equation 5 to zero:

$$0 = g_{\mathrm{L,E}}(E_L - \overline{v}_\lambda) + \overline{g}_\lambda(E_e - \overline{v}_\lambda), \quad \overline{v}_\lambda = \frac{g_{\mathrm{L,E}}E_L + \overline{g}_\lambda E_e}{g_{\mathrm{L,E}} + \overline{g}_\lambda}. \quad \text{(A3)}$$

Here $\overline{v}_\lambda$ is the deterministic mean-conductance operating-point voltage, not the exact expectation of the stochastic voltage; $g_{\mathrm{L,E}}$ is leak conductance; and E_L and E_e are leak and AMPA reversal potentials. Other symbols follow Equation A2.

A.2 Local synapse-plus-membrane response

Write each signal as its operating point plus a small fluctuation:

$$g_i = \overline{g}_\lambda + \delta g_i, \quad v_i = \overline{v}_\lambda + \delta v_i, \quad s_i = \frac{\lambda}{1000} + \delta s_i. \quad \text{(A4)}$$

Here δg_i, δv_i, and δs_i are complete conductance, voltage, and input perturbation variables around the means defined in Equations A2 and A3.

To derive Equation A5, substitute the conductance and input decompositions from Equation A4 into the synapse equation, Equation A1:

$$\frac{d(\overline{g}_\lambda + \delta g_i)}{dt} = -\frac{\overline{g}_\lambda + \delta g_i}{\tau_{\mathrm{AMPA}}} + w_{\mathrm{probe}}\left(\frac{\lambda}{1000} + \delta s_i\right).$$

The operating point is stationary, so its time derivative is zero. Expanding the right-hand side then gives

$$\frac{d\delta g_i}{dt} = \left(-\frac{\overline{g}_\lambda}{\tau_{\mathrm{AMPA}}} + w_{\mathrm{probe}}\frac{\lambda}{1000}\right) - \frac{\delta g_i}{\tau_{\mathrm{AMPA}}} + w_{\mathrm{probe}}\delta s_i.$$

The parenthesized stationary terms cancel by Equation A2. The remaining perturbation dynamics are

$$\frac{d\delta g_i}{dt} = -\frac{\delta g_i}{\tau_{\mathrm{AMPA}}} + w_{\mathrm{probe}}\delta s_i. \quad \text{(A5)}$$

To derive Equation A6, substitute the conductance and voltage decompositions from Equation A4 into the membrane equation, Equation 5:

$$C_E\frac{d(\overline{v}_\lambda + \delta v_i)}{dt} = g_{\mathrm{L,E}}(E_L - \overline{v}_\lambda - \delta v_i) + (\overline{g}_\lambda + \delta g_i)(E_e - \overline{v}_\lambda - \delta v_i).$$

As above, the deterministic operating point has zero time derivative. Expanding and grouping the right-hand side gives

$$C_E\frac{d\delta v_i}{dt} = \big(g_{\mathrm{L,E}}(E_L - \overline{v}_\lambda) + \overline{g}_\lambda(E_e - \overline{v}_\lambda)\big) - \big(g_{\mathrm{L,E}} + \overline{g}_\lambda\big)\delta v_i + (E_e - \overline{v}_\lambda)\delta g_i - \delta g_i \delta v_i.$$

The first parenthesized group is zero by Equation A3. The final product is second order in the small perturbations and is omitted by the local linear approximation. This leaves

$$C_E\frac{d\delta v_i}{dt} = -\big(g_{\mathrm{L,E}} + \overline{g}_\lambda\big)\delta v_i + (E_e - \overline{v}_\lambda)\delta g_i. \quad \text{(A6)}$$

Here C_E is excitatory-cell capacitance. All other symbols follow Equations A1–A4.

To derive Equation A7, take the Fourier transform of Equation A5. The derivative becomes $i\omega$ times the transformed signal:

$$i\omega\delta g_{i(\omega)} = -\frac{\delta g_{i(\omega)}}{\tau_{\mathrm{AMPA}}} + w_{\mathrm{probe}}\delta s_{i(\omega)}.$$

Move both conductance terms to the left, then divide by their common factor:

$$\left(i\omega + \frac{1}{\tau_{\mathrm{AMPA}}}\right)\delta g_{i(\omega)} = w_{\mathrm{probe}}\delta s_{i(\omega)},$$

$$\delta g_{i(\omega)} = \frac{w_{\mathrm{probe}}}{i\omega + \frac{1}{\tau_{\mathrm{AMPA}}}}\delta s_{i(\omega)}. \quad \text{(A7)}$$

Equation A8 follows in the same way from Equation A6. Fourier transformation first gives

$$i\omega C_E\delta v_{i(\omega)} = -\big(g_{\mathrm{L,E}} + \overline{g}_\lambda\big)\delta v_{i(\omega)} + (E_e - \overline{v}_\lambda)\delta g_{i(\omega)}.$$

Collect the voltage terms on the left:

$$\big(i\omega C_E + g_{\mathrm{L,E}} + \overline{g}_\lambda\big)\delta v_{i(\omega)} = (E_e - \overline{v}_\lambda)\delta g_{i(\omega)}.$$

Dividing by the coefficient of $\delta v_i(\omega)$ gives

$$\delta v_{i(\omega)} = \frac{E_e - \overline{v}_\lambda}{i\omega C_E + g_{\mathrm{L,E}} + \overline{g}_\lambda}\delta g_{i(\omega)}. \quad \text{(A8)}$$

Here ω is angular frequency, i is the imaginary unit, and each argument ω denotes a Fourier-domain signal.

Equations A5–A8 form a **linear time-invariant (LTI)** system after linearization about the operating point λ. Because the system has been linearized about the operating point, it is locally linear and time-invariant, so the standard transfer-function formalism applies. The canonical frequency-domain input–output relationship is

$$\delta v_{i(\omega)} = G_{\lambda(\omega)}\delta s_{i(\omega)}.$$

Here $G_\lambda(\omega)$ is the local transfer function relating input spike-train perturbations to membrane-voltage perturbations. For non-zero input, the input–output relationship can be rearranged as

$$G_{\lambda(\omega)} = \frac{\delta v_{i(\omega)}}{\delta s_{i(\omega)}}.$$

Thus the ratio follows from the standard input–output relationship rather than serving as an independent definition. To derive $G_\lambda(\omega)$, start from Equation A7:

$$\delta g_{i(\omega)} = \frac{w_{\mathrm{probe}}}{i\omega + \frac{1}{\tau_{\mathrm{AMPA}}}}\delta s_{i(\omega)}.$$

Substitute this expression for $\delta g_i(\omega)$ into Equation A8:

$$\delta v_{i(\omega)} = \frac{E_e - \overline{v}_\lambda}{i\omega C_E + g_{\mathrm{L,E}} + \overline{g}_\lambda}\left(\frac{w_{\mathrm{probe}}}{i\omega + \frac{1}{\tau_{\mathrm{AMPA}}}}\delta s_{i(\omega)}\right).$$

Reorder the scalar factors and factor out $\delta s_i(\omega)$:

$$\delta v_{i(\omega)} = \left(\frac{w_{\text{probe}}}{i\omega + \frac{1}{\tau_{\text{AMPA}}}} \cdot \frac{E_e - \overline{v}_\lambda}{i\omega C_E + g_{\text{L,E}} + \overline{g}_\lambda} \right) \delta s_{i(\omega)}.$$

Comparing this result with the canonical linear time-invariant relationship identifies the coefficient multiplying $\delta s_i(\omega)$ as the transfer function:

$$G_{\lambda(\omega)} = \frac{w_{\text{probe}}}{i\omega + \frac{1}{\tau_{\text{AMPA}}}} \cdot \frac{E_e - \overline{v}_\lambda}{i\omega C_E + g_{\text{L,E}} + \overline{g}_\lambda}. \quad \text{(A9)}$$

Equation A9 is therefore the synaptic filter multiplied by the membrane filter, not an arbitrarily chosen ratio. Its λ-dependence captures the change in membrane gain and effective time constant with mean conductance.

A.3 Finite-window averaging

Start from the baseline-subtracted feature in Equation 7:

$$z_i = \frac{1}{T} \int_0^T \left(v_{i(t)} - E_L \right) \mathrm{d}t.$$

Within the stationary linearized model used in this appendix, substitute the voltage decomposition from Equation A4:

$$z_i = \frac{1}{T} \int_0^T \left(\overline{v}_\lambda + \delta v_{i(t)} - E_L \right) \mathrm{d}t.$$

The operating-point voltage and leak reversal potential are constant in time, so their integral is their difference multiplied by T. Dividing by T gives

$$z_i = \overline{v}_\lambda - E_L + \frac{1}{T} \int_0^T \delta v_{i(t)} \, \mathrm{d}t.$$

Define the operating-point feature and its perturbation by

$$\overline{z}_\lambda = \overline{v}_\lambda - E_L, \quad \delta z_i = z_i - \overline{z}_\lambda.$$

Subtracting the operating-point feature from the expanded expression for z_i cancels the constant term and leaves

$$\delta z_i = \frac{1}{T} \int_0^T \delta v_{i(t)} \, \mathrm{d}t. \quad \text{(A10)}$$

Thus δz_i is the fluctuation around the operating-point feature, obtained by averaging the voltage perturbation over the 200 ms presentation duration T. To derive the frequency response, represent the averaging operation as a causal rectangular kernel $a_T(u)$, where

$$a_{T(u)} = \frac{1}{T}, \quad 0 \le u \le T,$$

and $a_T(u)$ is zero outside that interval. Applying this kernel as a moving-average filter gives

$$\delta z_{i(t)} = \int_{-\infty}^{\infty} a_{T(u)} \delta v_{i(t-u)} \, \mathrm{d}u = \frac{1}{T} \int_0^T \delta v_{i(t-u)} \, \mathrm{d}u.$$

At the end of the presentation, set t equal to T. To put the resulting integral into the same form as Equation A10, introduce the new variable

$$t' = T - u, \quad \mathrm{d}t' = -\,\mathrm{d}u.$$

When u is zero, t' is T; when u is T, t' is zero. The substitution therefore reverses the integration limits. Its minus sign reverses them back:

$$\delta z_{i(T)} = \frac{1}{T}\int_0^T \delta v_{i(T-u)}\,\mathrm{d}u = -\frac{1}{T}\int_T^0 \delta v_{i(t')}\,\mathrm{d}t' = \frac{1}{T}\int_0^T \delta v_{i(t')}\,\mathrm{d}t'.$$

The name of an integration variable has no effect on the value, so replacing t' with t recovers Equation A10 exactly.

For any time-domain function f(u), use the Fourier-transform convention

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} f(u)\exp(-i\omega u)\,\mathrm{d}u.$$

Here hat(f) (ω) is the Fourier transform of f(u). Applying this definition with f(u) equal to the rectangular kernel a_T (u), and naming the result A_T (ω), gives

$$A_{T(\omega)} = \int_{-\infty}^{\infty} a_{T(u)}\exp(-i\omega u)\,\mathrm{d}u = \frac{1}{T}\int_0^T \exp(-i\omega u)\,\mathrm{d}u.$$

Evaluating the integral and rearranging gives

$$A_{T(\omega)} = \frac{\exp(-i\omega T) - 1}{-i\omega T} = \frac{1 - \exp(-i\omega T)}{i\omega T}.$$

Thus a_T (u) and A_T (ω) are not alternative definitions: a_T is the time-domain rectangular averaging kernel, whose height 1/T makes its area one, whereas A_T is that kernel's Fourier transform and is the frequency response stated in Equation 14.

The identity $1 - \exp(-ix) = 2i\exp(-i\frac{x}{2})\sin(\frac{x}{2})$, with $x = \omega T$, gives the equivalent form

$$A_{T(\omega)} = \exp\left(-i\omega\frac{T}{2}\right)\frac{\sin\left(\omega\frac{T}{2}\right)}{\omega\frac{T}{2}}.$$

The exponential has unit magnitude and contributes only the phase delay of a window centred at T/2. The magnitude is therefore the sinc-shaped response

$$\left|A_{T(\omega)}\right| = \left|\frac{\sin\left(\omega\frac{T}{2}\right)}{\omega\frac{T}{2}}\right|.$$

In this experiment T is measured in ms and physical frequency f is measured in Hz, so $\omega = 2\pi\frac{f}{1000}$ rad/ms. Consequently,

$$\left|A_{T\left(2\pi\frac{f}{1000}\right)}\right| = \left|\frac{\sin\left(\pi f\frac{T}{1000}\right)}{\pi f\frac{T}{1000}}\right|.$$

At zero frequency this expression appears as 0/0, but its continuous limit is one, as expected: averaging does not change a constant signal. Its zeros occur at $f = 1000\frac{m}{T}$ Hz for nonzero integers m; for T = 200 ms, the first zero is therefore 5 Hz and subsequent zeros occur every

5 Hz. These zeros and intervening sidelobes produce the comb-like shape of Figure 4, Panel B.

By the convolution theorem, the averaging filter multiplies the voltage spectrum:

$$\delta z_{i(\omega)} = A_{T(\omega)} \delta v_{i(\omega)}.$$

Substitute the input–voltage relation from Equation A9:

$$\delta z_{i(\omega)} = A_{T(\omega)} G_{\lambda(\omega)} \delta s_{i(\omega)}.$$

Define the complete local input–feature response by the canonical linear time-invariant relationship

$$\delta z_{i(\omega)} = H_{\lambda(\omega)} \delta s_{i(\omega)}.$$

Comparing the coefficients multiplying δs_i (ω) identifies H_λ (ω). The two components of Equation A11 are therefore

$$A_{T(\omega)} = \frac{1 - \exp(-i\omega T)}{i\omega T}, \quad H_{\lambda(\omega)} = A_{T(\omega)} G_{\lambda(\omega)}. \quad \text{(A11)}$$

Here A_T (ω) is the averaging-window response, exp is the exponential function, and H_λ (ω) is the complete response from input spike-train perturbation to averaged-voltage perturbation.

For the Bode comparison, define the common reference $G_{\text{ref}} = \left|G_{\lambda_{\text{low}}}(0)\right|$, where $\lambda_{\text{low}} = 0.25$ spikes/s. The two plotted magnitudes are

$$B_{G(\omega)} = 20 \log_{10}\left(\frac{\left|G_{\lambda(\omega)}\right|}{G_{\text{ref}}}\right), \quad B_{H(\omega)} = 20 \log_{10}\left(\frac{\left|H_{\lambda(\omega)}\right|}{G_{\text{ref}}}\right). \quad \text{(A12)}$$

Here B_G and B_H are magnitudes in decibels; | · | is complex magnitude; and $\log_{10}$ is the base-ten logarithm. Figure 4 plots B_G in Panel A and B_H in Panel B. Because A_T (0) = 1, the low-drive DC values of G_λ and H_λ share this reference.

A.4 Poisson-noise variance

In timestep n, let $S_n \in \{0, 1\}$ indicate whether a spike occurred. For rate λ and timestep Δt, its spike probability and centred value are

$$p = \lambda\Delta\frac{t}{1000}, \quad \delta S_n = S_n - p.$$

A no-spike bin therefore has $\delta S_n = -p$, while a spike bin has $\delta S_n = 1 - p$; centring does not leave the no-spike bins at zero. For example, at λ = 200 spikes/s with 1 ms bins, p = 0.2 and the centred values are −0.2 and 0.8. With exp080′s 0.1 ms timestep, p = 0.02 and they are −0.02 and 0.98. In both cases their probability-weighted mean is zero.

Autocovariance measures whether these deviations from the mean tend to occur together at two different times. For discrete lag k it is $E[\delta S_n \delta S_{n+k}]$. Independent Poisson bins give zero covariance for $k \neq 0$; at $k = 0$, a bin is compared with itself and the covariance is its variance $p(1 - p)$.

The analytical model takes the continuous-time limit. It represents the event train from Equation A4 as $s_{i(t)} = \sum_j \delta(t - t_j)$, a sum of Dirac impulses rather than a series of finite ones and zeros, and defines the centred fluctuation $\delta s_{i(t)} = s_{i(t)} - \frac{\lambda}{1000}$. As Δt tends to zero, the discrete zero-lag variance becomes a Dirac impulse. At continuous lag u, the autocovariance is

$$R_{\delta s}(u) = E\left[\delta s_{i(t)} \delta s_{i(t+u)}\right] = \frac{\lambda}{1000} \delta(u).$$

Here E denotes an average over repeated Poisson realizations. A positive autocovariance would mean that above-mean deviations at one time tend to accompany above-mean deviations at the displaced time; a negative value would indicate opposing deviations. For a Poisson process, counts in disjoint intervals are independent, so this covariance is zero whenever $u \neq 0$. At zero lag, an event coincides with itself, producing the Dirac impulse δ(u). Its coefficient λ/1000 is the mean event count per ms and is the area of that impulse, not its finite height.

By the Wiener–Khinchin relation, the input power spectral density is the Fourier transform of the autocovariance. The Fourier transform of δ(u) is one at every angular frequency, so the Poisson input has the constant, or white, two-sided spectrum

$$S_{\text{in}(\omega)} = \frac{\lambda}{1000}. \quad \text{(A13)}$$

Here S_{in} (ω) is input power per unit angular frequency and λ is mean spike rate in spikes/s; division by 1000 expresses the point-process rate per millisecond. Passing this noise through Equation A11 gives

$$S_{z(\omega)} = \left|H_{\lambda(\omega)}\right|^2 S_{\text{in}(\omega)}. \quad \text{(A14)}$$

Here S_z (ω) is predicted feature-noise power and $|H_\lambda$ (ω)$|^2$ is the transmitted fraction of input-noise power. Under the two-sided angular-frequency convention, total predicted variance is

$$\text{Var}_{\text{linear}(z)} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left|H_{\lambda(\omega)}\right|^2 S_{\text{in}(\omega)} \, \mathrm{d}\omega. \quad \text{(A15)}$$

Here $\text{Var}_{\text{linear}}$ (z) is predicted feature variance and π is the circle constant; the remaining symbols follow Equations A11–A14. This result is exact for the stated linear stationary model, but only approximate for the finite, discrete, conductance-dependent probe. It is retained as an analytical diagnostic rather than as the generator of ANN inputs.